

(Basic)

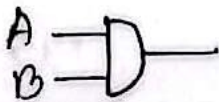
AND (·)
OR (+)
NOT (-)

(Universal)

NAND
NOR

(Exclusive)

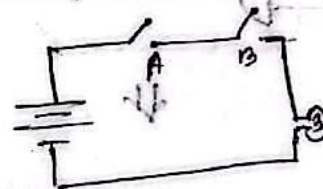
X-OR
X-NOR

(*) AND $\rightarrow$  A.B

Truth table: (AND)

Input		Output
A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

Circuit (AND)



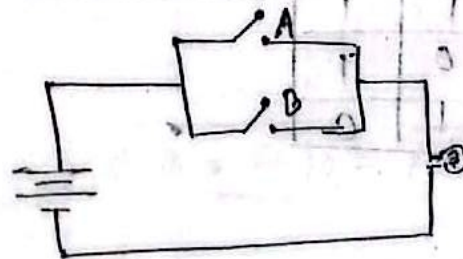
Output 1 when all input is 1.

(*) OR $\rightarrow$  A+B

Truth table: (OR)

Input		Output
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Circuit: (OR)



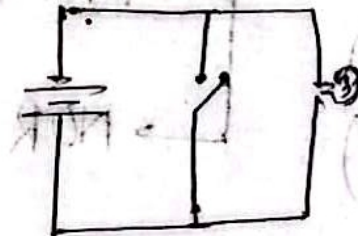
Output 0 when all input is 0.

(*) NOT $\rightarrow A \rightarrow \bar{A}$

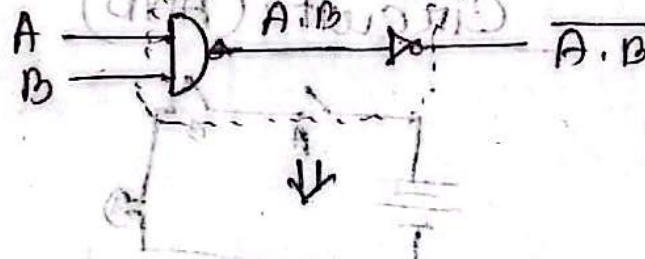
Truth table (NOT)

Input	Output
A	$\bar{A}$
0	1
1	0

Circuit (NOT)



(*) NAND (AND + NOT)



Input	Output
A	$\bar{A \cdot B}$
0	1
0	1
0	1
1	0

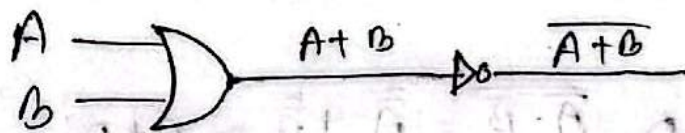
Truth table

in		Out
A	B	$\bar{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

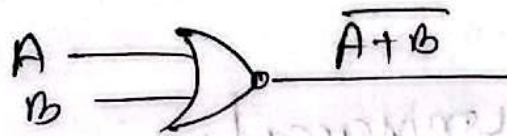


Input	Output
A	$\bar{A \cdot B}$
0	1
1	0

* NOR (OR + NOT)



⇓



Truth-table:

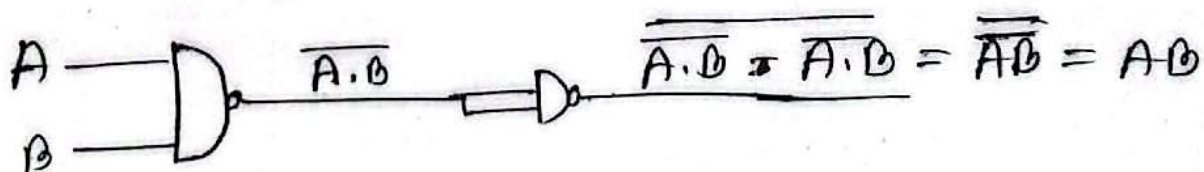
in		out	
A	B	$A+B$	$\overline{A+B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

(*) NAND $\rightarrow$ AND, OR, NOT

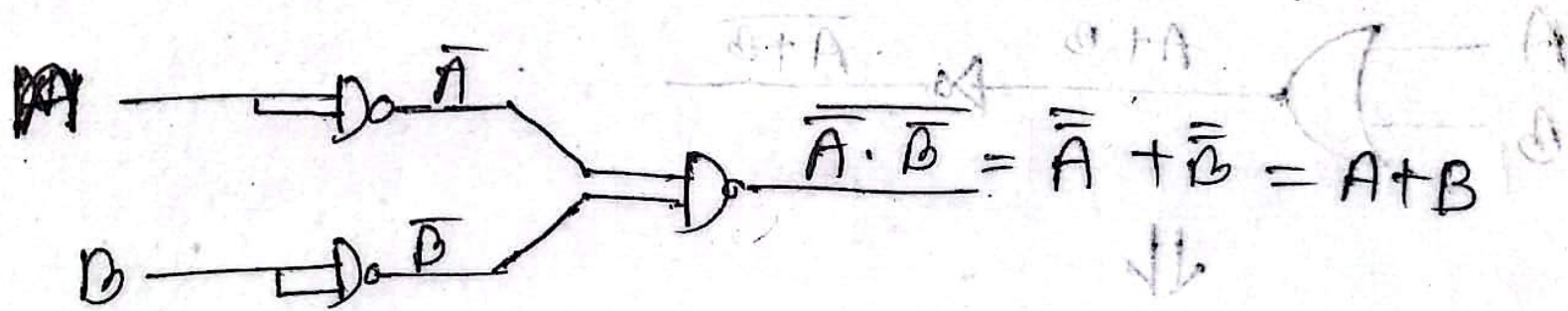
• NAND TO NOT:



• NAND TO AND:



NAND TO OR:



So, NAND is universal.

HW: Universality of NOR (NAND se
 kaise banayega)

A	B	A + B	A · B
0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	1